

DECISION TREE:

- Supervised learning algorithm

Classification Regression

- Recursively splitting the dataset into subsets on the best feature that maximizes purity (classification) or minimizes error (regression).

BEST FEATURE

• Reg → continuous (IV → num; DV → num)

Class →

MATHEMATICAL FOUNDATION

A DT select splits using impurity measures.

(A) ENTROPY:

$$\text{Entropy}(S) = - \sum_{i=1}^c p_i \cdot \log_2(p_i)$$

p = proportion

c = no. of class

if Ent(S) = 0

Pure node

if Ent(S) = 1

Max. impurity

$$\log_2(n) = \frac{\log(n)}{\log(2)}$$

eg: 5 boys, 9 girls

$$\rightarrow - \left[\frac{5}{14} \log_2\left(\frac{5}{14}\right) + \frac{9}{14} \log_2\left(\frac{9}{14}\right) \right]$$

(B) INFORMATION GAIN

$$IG(S,A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

A = Feature

S_v = subset after split

Feature

(C) GINI INDEX

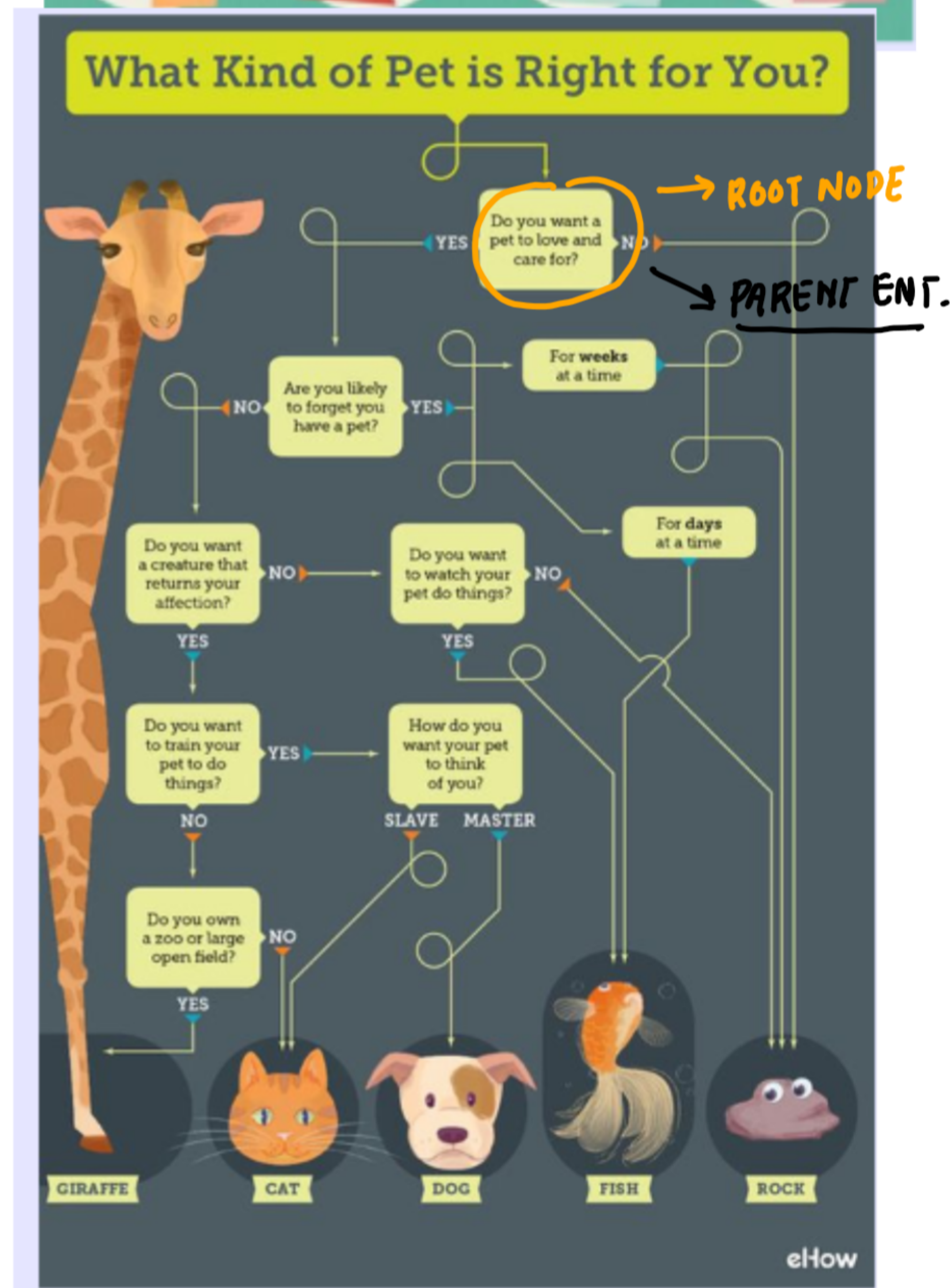
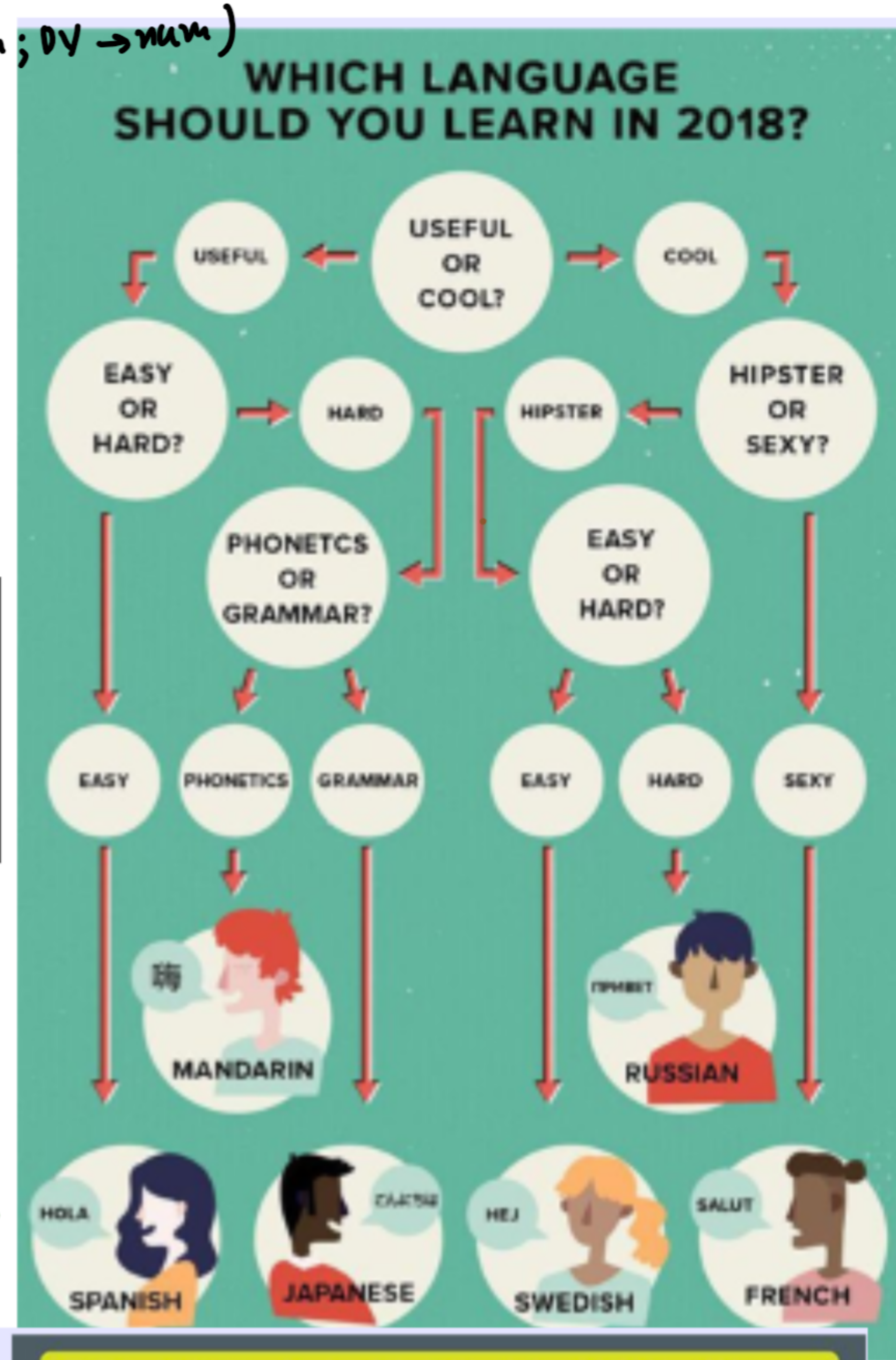
$$\text{Gini}(S) = 1 - \sum_{i=1}^c p_i^2$$

Gini(S) = 0

Pure

Gini(S) = 0.5

Max. impurity.



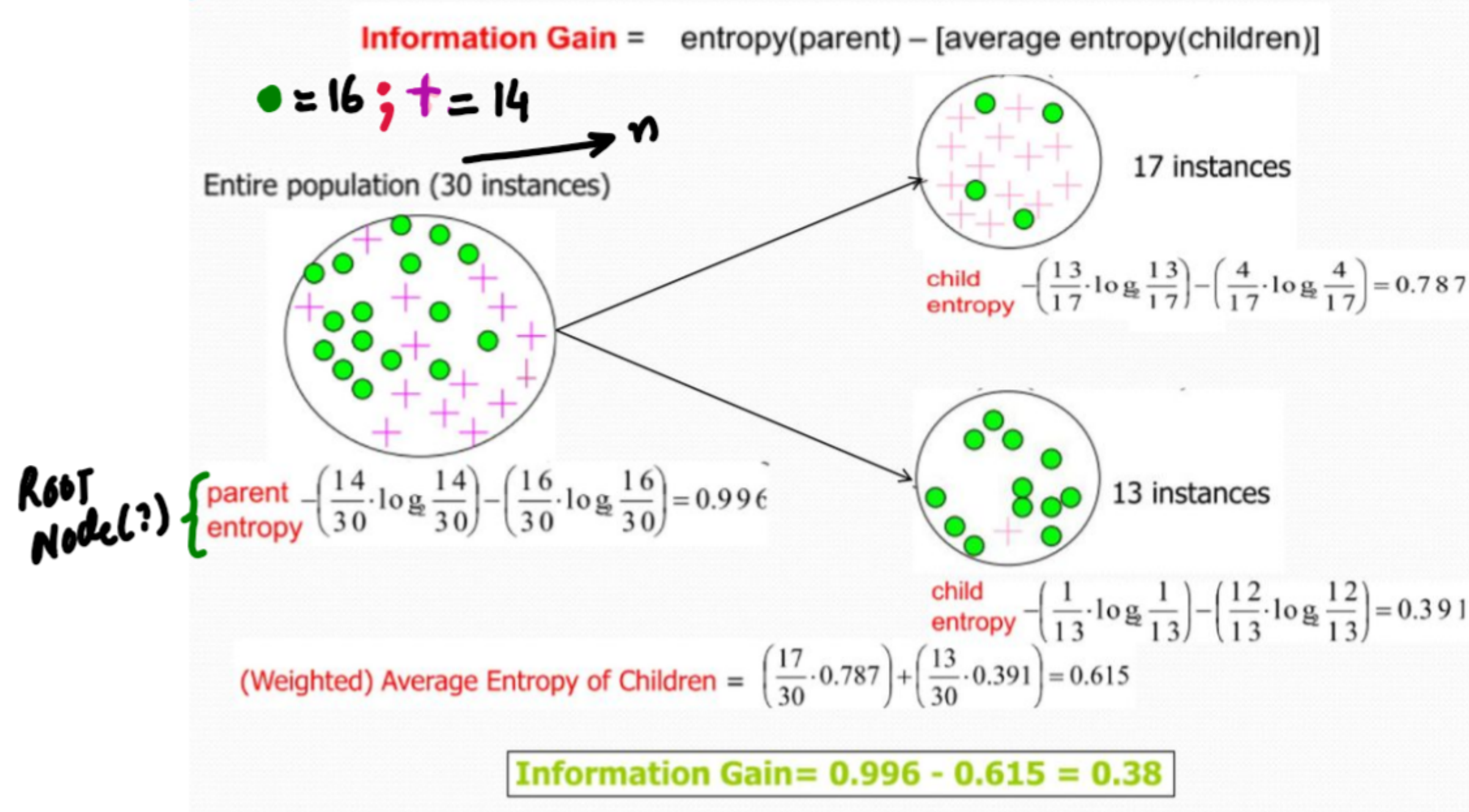
Exp: 1

Consider the following data, where the Y label is whether or not the child goes out to play.

Day	F_1 Weather	F_2 Temperature	F_3 Humidity	F_4 Wind	Y Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Exp: 2

Information gain calculation example



STEP-I: Entropy (S) $\rightarrow$ related to output (y)
 $n=10$
 $Y_{ES}=5 ; N_{O}=5$
 y is classification

$$P(YES) = P_1 = \frac{5}{10}$$

$$P(NO) = P_2 = \frac{5}{10}$$

$$\text{Entropy}(S) = - \left[\frac{5}{10} \log_2 \left(\frac{5}{10} \right) + \frac{5}{10} \log_2 \left(\frac{5}{10} \right) \right]$$

$$= - \left[(0.5 \times -1) + (0.5 \times -1) \right]$$

$$= - \left[-0.5 - 0.5 \right]$$

$$\boxed{\text{Entropy}(S) = 1}$$

for ref: $\log_2 \left(\frac{5}{10} \right) = \frac{\log \left(\frac{5}{10} \right)}{\log(2)}$
 $= -1$

STEP-II: INFORMATION GAIN FOR FEATURE-I (F_1)

F_1 WEATHER

$n=10$

S_v Sunny=3 ; Cloudy=3 ; Rainy=4

$Y=1, N=2$ $Y=3, N=0$ $Y=1, N=3$

$$Y=1, N=2$$

$$P_1 = \frac{1}{3} ; P_2 = \frac{2}{3}$$

$$\textcircled{1} - \left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3} \right) = 0.917$$

$$Y=3, N=0$$

$$P_1 = 1 ; P_2 = 0$$

$$\textcircled{2} - \left(\frac{3}{3} \log_2 \frac{3}{3} + \frac{0}{3} \log_2 \frac{0}{3} \right) = 0$$

$$Y=1, N=3$$

$$P_1 = \frac{1}{4} ; P_2 = \frac{3}{4}$$

$$\textcircled{3} - \left(\frac{1}{4} \log_2 \frac{1}{4} + \frac{3}{4} \log_2 \frac{3}{4} \right) = 0.81$$

$$\text{Ent}(S) = \sum \frac{|S_v|}{|S|} \cdot \text{Ent}(S)$$

$$IG = 1 - \left[\frac{3}{10} (0.917) + \frac{3}{10} (0) + \frac{4}{10} (0.81) \right] \cdot 1$$

$$= 1 - 0.59$$

$$IG = 0.41$$

TASK=

Find IG for F_2, F_3, F_4 !

Compare which feature has more IG among others?